3(b)	The R–C smoothing filter in 50 Hz mains operated half-wave rectifier circuit consists of $R_1 = 100\ \Omega$ and $C_2 = 1{,}000mF$. If 1 V of ripple appears at the input of the circuit, determine the amount of ripple appearing at the output.	04	L3	CO3
4(a)	Illustrate the operation of a BJT transistor in common base configuration using a neat circuit diagram, and discuss its input and output characteristics.	07	L2	CO3
4(b)	In a BJT amplifier configuration, the current amplification factor α is 0.9. If the emitter current is 1mA, determine the value of base current and β.	03	L3	CO3

DAYANANDA SAGAR UNIVERSITY

SCHOOL OF ENGINEERING

USN No: ENG25CS0453

B. Tech II Semester End Examinations – May 2026

Course Title: Physics for Computer Science Cluster — **Course Code:** 25EN1115

Duration: 02 Hours 30 Minutes — **Date:** 21-05-2026

Time: 01:00 PM to 03:30 PM — **Max Marks:** 80

Note:
1. **Answer all Five Full Questions**
2. **Each Full Question carries 16 Marks**
3. **Draw neat sketches wherever necessary**
4. **Missing Data may be suitably assumed**

Q. No.	Question	Marks	BTL	COs
1.a.	Define wave function. Explain its physical significance and the conditions for a valid wave function.	04	L2	CO1
1.b.	An electron is bound in a one-dimensional infinite well of 2 x 10^{-12} m. Find the difference in energy values of the first two excited states. (h= 6.626 X 10^{-34} J-s)	04	L2	CO1
1.c.	Solve Schrödinger's equation for a particle in a 1D infinite potential well and show that energy eigenvalues are discrete.	08	L3	CO1
2.a.	Define a qubit and explain the quantum entanglement?	04	L2	CO2
2.b.	What is quantum parallelism?	04	L2	CO2
2.c.	Write the matrix representation of the Pauli-X gate and explain its action on a qubit.	03	L3	CO2
2.d.	Describe the working of Quantum Key Distribution (QKD).	05	L3	CO2
3.a.	Differentiate between Spontaneous and Stimulated emission.	03	L2	CO3
3.b.	Describe the working of an Nd:YAG laser using energy level diagram.	07	L3	CO3
3.c.	Discuss the different types of losses in optical fibers.	06	L3	CO3
4.a.	Derive an expression for the Hall Voltage (V_H)	05	L3	CO4
4.b.	At 50 °C, calculate the probability that an energy level 0.2 eV above the Fermi level is occupied by an electron. (K_B = 8.617 × 10^{-5} eV/K).	03	L2	CO4